ML Model Evaluation & Explainability Report

Generated on 2026-09-23 13:00:30 | Session ID: d2111706-bee4-4382-8f5b-1b29d13d5ea1

1. Dataset & Problem Detection Summary

Dataset File:	loan_data.csv	**Total Rows:**	5004
Target Column:	Loan_Status	**Total Columns:**	16
Problem Type:	Classification	**Test Split:**	20%

2. Preprocessing & Validation Protocol

Numerical Imputer	Median	**Categorical Encoding**	Onehot
Categorical Imputer	Most_frequent	**Feature Scaling**	Standard
Random State	42	**CV Protocol**	2 Folds

3. Model Performance Comparison

Rank	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Train Time (s)
1	Decision Tree	0.8192	0.6269	0.6747	0.6499	0.7709	0.14

4. Recommended Champion Model

Recommended Model: Decision Tree
Criterion: Highest F1 Score (0.6499).
Rationale: Superior performance and fold consistency on held-out validation data.